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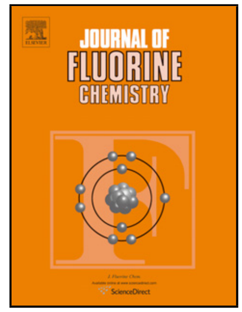

Borolan-2-yl involving anion acceptors for organic liquid electrolyte-based fluoride shuttle batteries

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Borolan-2-yl involving anion acceptors for organic liquid electrolyte-based fluoride shuttle batteries

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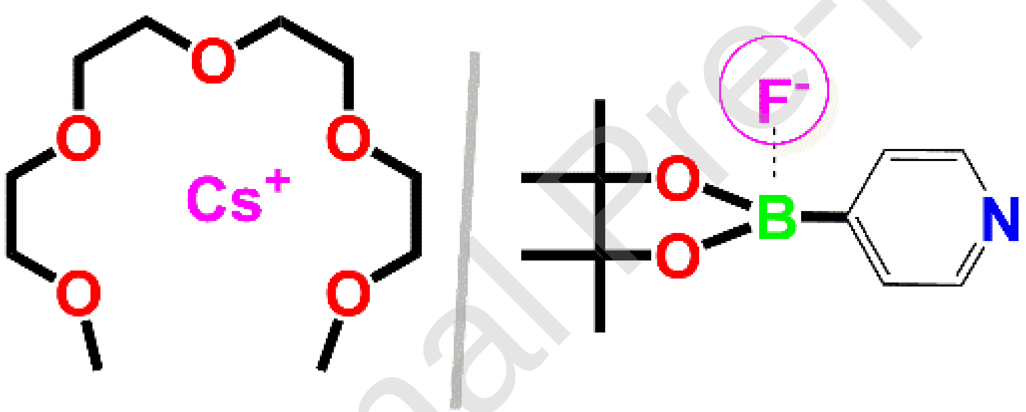
Graphical abstract



possible interactions in the Py/CsF/G4 electrolyte

Highlights

- Two boron-based anion acceptors (AAs; DiOB-Py , DiOB-An) were introduced.
- The effects of the acidity strength of AAs in G4 containing CsF investigated.
- Then, electrochemical compatibility of BiF_3 were investigated for FSBs.
- DiOB-Py imparted relatively high fluoride ion conductivity and CsF solubility.
- Therefore, DiOB-Py allowed enhanced FSB performance.

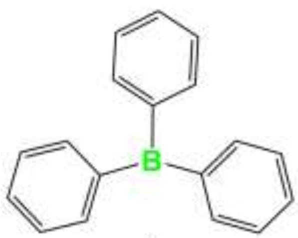
Abstract

Boron-based anion acceptors (AAs) have been used to dissociate MF salt (CsF) in organic liquid electrolyte-based fluoride shuttle batteries (FSBs). Here, two boron-based compounds, 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine (DiOB-Py) and *N*,*N*-diethyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline) (DiOB-An), were examined as electrolyte additives for G4 containing CsF. DiOB-Py and DiOB-An are structurally similar except that the former is functionalized with pyridine whereas the latter with aniline. Since pyridine in DiOB-Py withdraws an electron pair on the phenyl ring via the resonance effect, the Lewis acidity of borate increases. On the other hand, since lone pair electrons of aniline are donated to the phenyl ring through the resonance effect, the acidity of borate in DiOB-An decreases. AAs are known to have a moderately strong fluoride affinity to provide sufficient dissolution of CsF in the electrolyte solvent. Thus, the effects of the acidity strength of the borates on the electrochemical compatibility of BiF_3 were investigated. DiOB-Py imparted relatively high fluoride ion conductivity and CsF solubility, thereby allowing enhanced FSB performance.

Keyword: Fluoride shuttle battery, Boron-based anion acceptors, organic liquid electrolyte,

1. Introduction

Organic liquid electrolyte-based fluoride shuttle batteries (FSBs) rely on boron-based anion acceptors (AA), such as triphenylborane (TPhB;1),**[1]** fluorobis(2,4,6-trimethylphenyl)borane (FBTMPhB; 2),**[2-4]** and triphenylboroxin (TPhBX; 3).**[5-7]** They can bind to fluoride anions through Lewis acid-base interaction, thereby achieving fluoride salt solubility in the electrolyte. Boron-based anion acceptors (AA) have been mainly used in conjunction with tetraglyme (G4) for the positive bismuth fluoride (BiF_3) electrode in FSBs. So far, CsF was used as a fluoride source, whereas boron-based AAs provided the dissociation of CsF through its affinity for fluoride anions. Compared to TPhBX, TPhB (with 0.45 M CsF) provided enhanced discharge and charge capacities and cycling performance.



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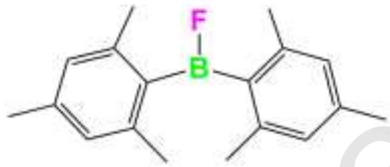

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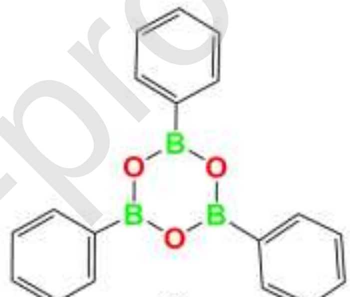

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In addition, increased salt concentration strongly influenced battery performance. The increase in CsF/AA ratio reduced the dissolution of the active material (Bi) in the electrolyte, especially during charging, thereby improving cycle performance.[5] Notably, AA must have a moderately strong fluoride affinity because an extremely strong fluoride affinity prevents fluoride anion of metal fluoride (MFx) from being drawn into the solution at a sufficient concentration during the discharge process (Eq. 1); thus, AA cannot release fluoride (from CsF). On the other hand, the low affinity of AA would cause insufficient dissolution of CsF.[8-10] In such cases, reversible conversion redox reaction processes[11-13] (Eqs. 1 and 2) cannot be successfully advanced. M and M′ represent different metals, and MFx and M′Fy represent their corresponding fluorides.

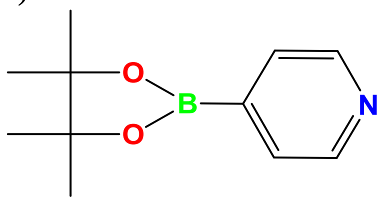
At the *positive* electrode: $xe^- + MF_x \rightarrow M + xF^-$ (1)

At the *negative* electrode: $\gamma F^- + M' \rightarrow M'F_\gamma + \gamma e^-$ (2)

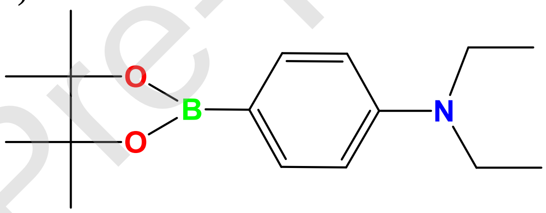
Based on the reported atomic absorption spectroscopy (AAS) results, one mole of AA dissolves nearly one mole of CsF. On the other hand, fluoride salt solubility can also be achieved without using AA in the presence of lithium bis(oxalato)borate (LiBOB).[14] Although fluoride shuttle-based redox reaction of BiF_3 was succeeded in LiBOB-based G4 electrolyte containing saturated CsF, Cs solubility was found to be 4.4 x 10^{-3} mol dm^{-3} in the presence of 0.25 mol dm^{-3} LiBOB. Compared to AA, Cs solubility is significantly low in the LiBOB-based G4 solution. Therefore, the presence of AAs has a critical role in FSB performance due to their ability to provide high solubility and diffusivity of fluoride anion complexes.[15]

In this study, two boron-based compounds with similar structures (borates) but different acidity strengths were examined as electrolyte additives for G4 containing CsF. The effects of the acidity strengths of the borates were investigated on BiF_3's electrochemical compatibility and thus FSB performance. Here, (borolan-2-yl)pyridine (DiOB-Py; 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine) and (borolan-2-yl)anniline (DiOB-An; N,N-diethyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline) (Scheme 1) derivatives have been introduced as AAs for organic liquid electrolyte-based FSBs. As can be seen in Scheme 1, the structural difference between the two structures is that DiOB-Py is functionalized with pyridine while DiOB-An with aniline.

a)



b)



Scheme 1 Structure of 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine (a) and *N,N*-diethyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline (b).

The potential effects of using DiOB-Py or DiOB-An as an electrolyte additive for G4 containing CsF on the electrochemical compatibility of the BiF_3 electrode were investigated by cyclic voltammetry (CV) and charge–discharge tests. The related discharge and charge reactions were confirmed by X-ray diffraction (XRD) analysis. In addition, dissolution probability of the active material after the discharge/charge process was investigated using atomic absorption spectroscopy (AAS).

2. Experimental

2.1 Electrode preparation and characterization

The cathode was prepared under inert conditions in a glove box (Miwa Manufacturing Co., Ltd.). Bismuth fluoride (Fluorochem Ltd.), acetylene black (AB) (Denka Black Li-100), and polyvinylidene difluoride (PVDF) (Kureha America KF1120) were used as an active material, carbon source, and cathode binder, respectively. The BiF_3/C nanocomposite-based active materials were prepared as described in the literature.[16, 17] Briefly, BiF_3 and AB (80:20 wt.%) were first mixed and milled for 1 hour at 1100 rpm; then, this mixture was dissolved in N-methylpyrrolidone (NMP) (Aldrich) to obtain a smooth (BiF_3/C)/C/PVDF slurry (60:25:15 wt.%) that was coated by doctor blade on an Al current collector (Nilaco Corporation). After kept for 2 h in the glove box, the mixture was pressed and dried overnight in a vacuum oven at 110°C. The crystal structures of the obtained BiF_3/C nanocomposite electrodes were analyzed using an XRD system (Rigaku Smartlab) with a Cu Kα (λ = 1.54 Å) radiation source. The XRD patterns were recorded in the 25−55° 2θ range.[14] All discharged and charged BiF_3 electrodes were packed into a pocket made by a transparent laminate sheet before they were transferred from the glove box for protection against air exposure.

2.2 Electrolyte preparation and characterization

The pyridine-based electrolyte was fabricated by the addition of saturated CsF (Tokyo Chemical Industry Co., Ltd.) as a fluoride salt to the G4 solvent (Kishida Chemical Co., Ltd.). In a glove box, 0.5 M DiOB-Py or DiOB-An (Aldrich) was added to the electrolyte solution. Then, the mixture was stirred at 60 ºC for 5 days. The resulting electrolyte solutions with saturated CsF concentration mentioned above was named Py/CsF/G4 and An/CsF/G4, respectively. The ionic conductivity was measured using a multipotentiostat (Biologic VMP300) over a frequency range $1–10^6$ Hz with an amplitude of 10 mV. AC impedance measurements was performed using the symmetric cell (Pt/electrolyte/Pt) to calculate the ionic conductivity of the liquid electrolytes.The electrochemical window of the electrolyte was evaluated by performing cyclic voltammetry (CV) at a scan rate of 0.1 mV s^{-1} with a three-electrode electrochemical cell (EC) (EC Frontier Co., Ltd.) comprising a platinum foil as a counter electrode, a carbon glassy (CG) as a working electrode with a diameter of 0.3 cm, and a silver rod immersed in acetonitrile with 0.1 M silver nitrate and 0.1 M tetraethylammonium perchlorate as a reference electrode (REF) (0.587 V vs standard hydrogen electrode (SHE)). CV was conducted on the prepared electrolytes at RT using the Biologic VMP300 multipotentiostat from – 4 V to 2 V and from 0 to 1.5 V (vs REF), respectively, at a sweep rate of 1 mV s^{-1}.

2.3 Electrochemical performance of BiF_3/C nanocomposite-based electrode with pyridine-based electrolyte

The discharge and charge capacities of the BiF_3/C nanocomposite-based electrode were measured using a three-electrode electrochemical cell consisting of a BiF_3/C composite electrode as a working electrode, a platinum mesh as a counter electrode, and a silver rod immersed in acetonitrile with 0.1 M silver nitrate and 0.1 M tetraethylammonium perchlorate as REF. The electrochemical cells were assembled in a glove box.

The discharge and charge measurements were performed at RT using a multipotentiostat (Biologic VMP300) from –1.8 to 0.0 V (vs REF) at 1/40 C, 1/20 C, and 1/10 C (for BiF_3 as an active material; 1 C = 302 mA g^{-1}). The specific capacity was derived from dividing the capacity by the active material mass.[18]

2.4 CsF/KF solubility in the electrolyte

The amount of Cs or Bi dissolved in the electrolyte was analyzed by ICP-MS Agilent 7700x.

3. Result and Discussion

In this work, (borolan-2-yl)pyridine (DiOB-Py; 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine) and (borolan-2-yl)anniline (DiOB-An; N,N-diethyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline) derivatives (Fig. 1) were used as a new type of AA for FSB. Therefore, their 0.5 M concentrations with saturated CsF in G4 were prepared. They were named as Py/CsF/G4 and An/CsF/G4, respectively. The CsF solubilities in Py/CsF/G4 and An/CsF/G4 determined by ICP-MS were found as 3,900 ppm (0.03 M) and 2,200 ppm (0.016 M), respectively (Table 1), indicating that the solubility of CsF in the pyridine-based G4 electrolyte is approximately 1.8 times higher than that in aniline-based G4 electrolyte.

Table 1 Solubility of CsF in the Py/CsF/G4 and An/CsF/G4 solvent systems, as determined by ICP-MS.

Solvent	Element	Concentration (ppm (μg/g))
Py/CsF/G4	Cs	3,900
An/CsF/G4	Cs	2,200

The ionic conductivity was found to be 2.71×10^{-5} S cm^{-1} for Py/CsF/G4 and 3.18×10^{-6} S cm^{-1} for An/CsF/G4 at 25°C (Table 2). The concentration and temperature dependence of their conductivities exhibited linear behaviors according to the Arrhenius equation, as shown in Fig. 1. For every increment in the temperature, the conductivities of Py/CsF/G4 and An/CsF/G4 increased a bit. Although the CsF solubility in An/CsF/G4 was 1.8 times lower, as confirmed by ICP-MS, this electrolyte solution exhibited nearly one tenth conductivity compared to Py/Cs/G4 at 25°C. At that moment, the exact reason behind the lower ionic conductivity of An/CsF/G4 compared to CsF solubility is currently unclear.

Table 2 Comparison between the temperature dependence of the conductivity (S cm^{-1}) of the Py/CsF/G4 and An/CsF/G4 solutions (DiOB-Py;, 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine, DiOB-An; N,N-diethyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline, G4: tetraglyme), Ea; Activation Energy).

	25℃	35℃	45℃	55℃	65 ℃	Ea (kJ mol^{-1})
Py/CsF/G4	2.71×10^{-5}	3.02×10^{-5}	3.29×10^{-5}	3.51×10^{-5}	3.69×10^{-5}	6.6
An/CsF/G4	3.18×10^{-6}	3.64×10^{-6}	4.12×10^{-6}	4.43×10^{-6}	4.70×10^{-6}	8.3

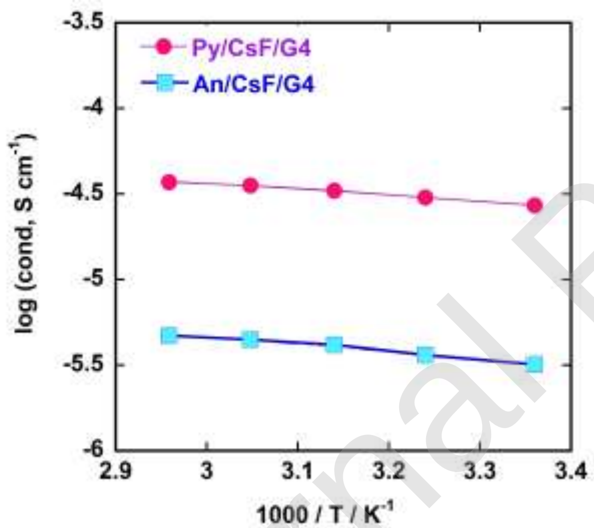


Figure 1 Temperature dependence of the conductivity of the electrolytes prepared using tetraglyme (G4), DiOB-Py: 4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine or DiOB-An; N,N-diethyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)aniline with saturated CsF.

It is well-known that fluoride anion affinity of boron-based AA increases with the number of electron-withdrawing groups.[19-21] Therefore, the acidity of borate in DiOB-Py increases as pyridine withdraws a pair of electrons on the phenyl ring through the resonance effect. On the other hand, since lone pair electrons of aniline are donated to the phenyl ring through the resonance effect, the acidity of borate in DiOB-An decreases. Therefore, CsF solubility and diffusivity of fluoride anion complexes are higher in Py/Cs/G4 since the acidity of borate in DiOB-Py is higher than that in DiOB-An. Obtaining higher conductivity in Py/Cs/G4 demonstrates the effect of the structure of the anion acceptor on the conductivity and confirms that DiOB-Py rather than DiOB-An enhances the dissolution of CsF in G4.

The electrochemical stability of the prepared electrolytes analyzed using CV (Fig.2a) showed that Py/CsF/G4 exhibited a reduction peak at a voltage below −2.8 V (vs. REF) and two oxidation peaks at a voltage above − 0.20 V (vs. REF) and 1.6 V. On the other hand, An/CsF/G4 (Fig. 2b) exhibited a reduction peak at a voltage below −3 V (vs. REF) and an oxidation peak at a voltage above − 0.20 V (vs. REF). To determine the oxidation stability, the CV measurement was started with the oxidation state (Fig. 2b, inset); only a single oxidation peak above 0.4 V (vs REF) for An/CsF/G4 was observed starting from the second cycle. These CV results show that Py/CsF/G4 and An/CsF/G4 have electrochemical stability between −2.8 V and −0.2 V and between −3 V and 0.4 V, respectively. The stability range of An/CsF/G4 (3.4 V) is wider than that of Py/CsF/G4 (2.6 V). On the other hand, two reference solutions containing 0.5 M DiOB-Py and 0.5 M DiOB-An without CsF were alternatively prepared and named as Py/G4 and An/G4, respectively. As can be seen from the graph (Fig 2a and 2b), they showed almost no oxidation and reduction peaks, indicating the lack of electrochemical activity.

a) b)

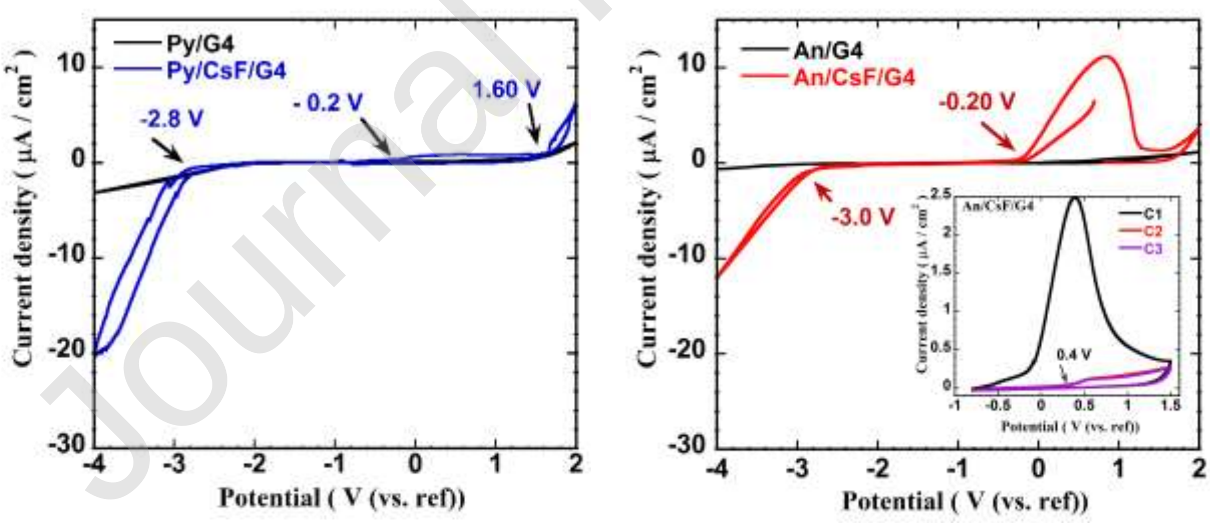


Figure 2 CV of the glassy carbon electrode in (a) Py/CsF/G4 and (b) An/CsF/G4 in the potential ranges from –4 to 2 V (vs reference electrode) at scan rate 0.1 mV s^{-1}. Insets is in the potential ranges from 0.0 to 1.5 V (vs reference electrode) at scan rate 0.1 mV s^{-1} (DiOB-Py:

4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine; G4: tetraglyme).

To investigate the effects of (borolan-2-yl)pyridine and (borolan-2-yl)anniline derivatives as AA on the electrochemical compatibility of BiF_3/C nanocomposite electrodes for organic liquid electrolyte-based FSB systems, the potential profiles of BiF_3 in the Py/CsF/G4 and An/CsF/G4 electrolytes were analyzed. The discharge and charge reactions of BiF_3 were hindered in An/CsF/G4 since no capacity values were obtained. This result can be explained by the low CsF dissolution and especially the low ionic conductivity of the An/CsF/G4 electrolyte.
On the other hand, the cycling ability of the BiF_3/C nanocomposite electrodes in Py/CsF/G4 was investigated in Fig. 3. Almost the full discharge and charge capacity of BiF_3 were obtained in Py/CsF/G4 electrolyte at C/40. The capacity obtained for the first discharge and charge is 425/263 mAh g^{-1} at a cut-off potential of –1.8 V and 0.0 V. The second and third discharge/charge capacities of BiF_3/C were almost the same (330/250 mAh g^{-1}).

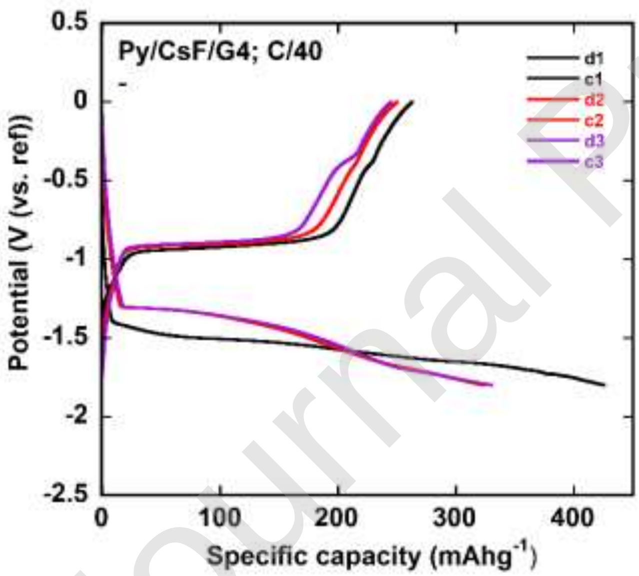


Figure 3 Some initial discharge–charge curves of the BiF_3/C nanocomposite electrodes, cycled at room temperature, and C/40 rate in Py/CsF/G4 (discharging cuto voltage: –1.8 V; charging cuto voltage: 0.0 V).

To identify the discharging process, the crystal structure of BiF_3 was analyzed after the end of the first discharge cycle for Py/CsF/G4 (Fig. 4). The XRD pattern obtained after the cut-off potential at –1.8 V at 1/40 C rate clearly shows Bi peaks; all diffraction peaks of cubic BiF_3 were completely disappeared below –1.8 V. This result confirms that the discharge capacity obtained in the first cycle is mainly due to the conversion of BiF_3 into metallic Bi. On the other hand, XRD pattern was investigated after the charge cut-off potential at 0.0 V. The presence of the peaks related to BiF_3 (cubic phase) indicates that the charging process was starting to occur. However, the presence of Bi peaks indicates that the charging process was not complete. This result is consistent with the reports on FSB using liquid-based electrolyte systems.

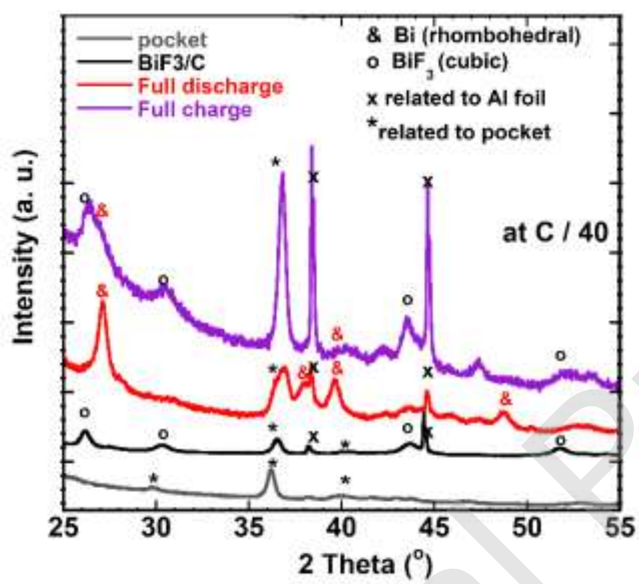


Figure 4 X-ray diffraction patterns of BiF_3/C in the pristine, fully discharged, and fully charged states for Py/CsF/G4 at C/40.

In liquid-based FSB studies, the charge-discharge capacity loss is mainly observed after the first discharge.[19] In contrast, although there is a slight decrease after the first discharge, the full capacity value can still be achieved in the second and third capacities, and then it is exceeded after the third run. In these studies, together with the low electronic conductivity and volume change problems of the BiF_3 positive electrode, the high reactivity of Bi (produced during the discharge process) toward the electrolyte was emphasized as an important reason. These limitations stated in the literature could be some of the reasons behind not reaching full capacity during the charging process.

Recently, the performance of the BiF_3/C electrode was investigated in $TPhBX_{0.5}$/sat_CsF/G4 electrolyte.[22] The $TPhBX_{0.5}$/sat_CsF/G4 system proposed by Konishi et al. contains saturated CsF and 0.5-M TPhBX in G4, whereas the Py/CsF/G4 system introduced here by Kucuk et al. contains saturated CsF and 0.5-M (borolan-2-yl)pyridine in G4. As seen, the difference of these two systems is the type of used anion acceptor. The performance of BiF_3 in Py/CsF/G4 was compared with the similar state-of-the-art electrolyte, $TPhBX_{0.5}$/sat_CsF/G4. In both studies, BiF_3 was pulverized by AB for 1 h at 1100 rpm by ball milling method and used as positive electrode. The obtained capacities of discharge and charge are about 400 and 210 mAh/g, respectively in the $TPhBX_{0.5}$/sat_CsF/G4 system. However, they are 425 and 260 mAh/g, respectively in the Py/CsF/G4 system. As seen, the first important improvement is on the obtained reversible capacity. The first charge capacity of BiF_3 in the Py/CsF/G4 system is higher than that of in the $TPhBX_{0.5}$/sat_CsF/G4 system. On the other hand, second and another important achievement is on the polarization. BiF_3 in the $TPhBX_{0.5}$/sat_CsF/G4 system could be characterized by a relatively high discharge/charge polarization and the amount of polarization varies with cycling in a similar manner to cycling stability. However, BiF_3 in the Py/CsF/G4 system showed a relatively low discharge/charge polarization and the amount of polarization almost did not change with the increasing cycling. It can be assumed that the discharge and charge plateaus mainly correspond to the conversion redox reaction of active materials (Eq. 1 and Eq. 2, respectively).During the discharging and also charging processes there are longer and stable plateau for redox reaction of BiF_3 in Py/CsF/G4 system compared to the reported work mentioned above, clearly indicating more effective redox reaction of BiF_3.

The performance of the BiF_3/C nanocomposite electrode in Py/CsF/G4 electrolyte was further investigated by increasing the C rate. The discharge and charge curves for BiF_3 at 1/20 and 1/10 C at a cutoff potential of –1.8 and 0.0 V (vs REF) for both C rates are shown in Fig. 5. The first discharge and charge capacities were, respectively, 228 and 180 mAh g^{-1} at 1/20 C and 236 and 160 mAh g^{-1} at 1/10 C. A notable decrease was observed in the initial discharge capacity for both 1/20 and 1/10 C rates compared to that at 1/40 C; this difference could be caused by IR losses due to the cell resistance and possible electrolyte degradation with increasing C rate during the discharge.

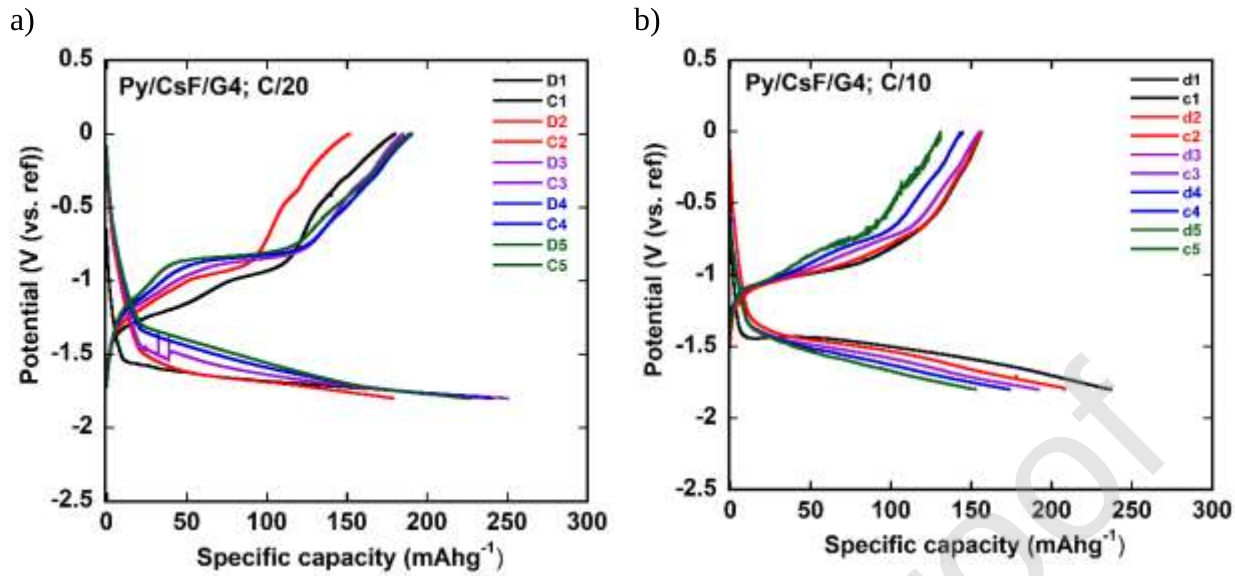


Figure 5 Some initial discharge–charge curves of the BiF_3/C nanocomposite electrodes, cycled at room temperature, and (a) for C/20 rate and (b) for C/10 rate in Py/CsF/G4 (discharging cuto□ voltage: –1.8 V; charging cuto□ voltage: 0.0 V).

The contribution of the electrochemical dissolution of the active material on the exceeding theoretical capacity was investigated by ICP-MS. Therefore, the amount of Bi concentration in the electrolyte (Py/CsF/G4) after fully discharged and charged states was examined (Table 3). As seen in Table 3, almost no Bi was detected in the electrolyte solutions in both fully discharged (0.1<) and charged states (0.1<). Moreover, to investigate the chemical dissolution of BiF_3 powder in Py/CsF/G4, Py/CsF/G4 electrolyte with saturated BiF_3 concentrations was prepared. The solubility of BiF_3 in the Py/CsF/G4 electrolyte was investigated using ICP-MS (Table 3). The concentrations of Bi were extremely low in Py/CsF/G4 (4.8 ppm), implying that even chemical dissolution did not occur. Since the electrochemical and chemical dissolution of the active material is extremely low or does not occur, the loss of the active material or side reactions between electrolyte and dissolved active material causing capacity reduction are unlikely.

It has been reported that the electrochemical reactions in the G4-based electrolyte (G4-AA) occur mainly through the dissolution-deposition mechanism.[23] Furthermore, it has been demonstrated that the battery reaction mechanism depends not only on the type of solvent used but also on the combination of electrolyte components.[23] That is, the contribution of predominant dissolution-deposition mechanism decreases when high salt concentration is used compared to AA in the G4 based electrolyte. In another study, it was found that when only 1% w/w LiBOB was added to the system (to G4/AA), the mechanism faces an interesting change and turn to the direct defluorination mechanism.[21] On the one hand, it is well known that in the case of the dissolution–deposition mechanism, active materials should first be dissolved in the electrolyte. Therefore, the ICP-MS results suggest that the reduction reaction for the pyridine-based electrolyte system may not depend on the dissolution–deposition mechanism or that its contribution is weak. Based on the ICP-MS results, we strongly believe that the direct defluorination mechanism has occurred more effectively in Py/CsF/G4.

Table 3 Solubility of Bi in the Py/CsF/G4 electrolyte system after full discharge and charging, as determined by ICP-MS

Solvent	Element	Concentration (ppm (μg/g))
Py/CsF/G4 (mixed with saturated BiF_3 powder)	Bi	4.8
Py/CsF/G4 after full discharge	Bi	0.1 <
Py/CsF/G4 after full charging	Bi	0.1 <

4. Conclusion

Two different boron-based compounds with different Lewis acidity strengths, (borolan-2-yl)pyridine (DiOB-Py) and (borolan-2-yl)anniline (DiOB-An), were employed as a new type AA for organic liquid electrolyte-based FSBs.

Although the CsF solubility in An/CsF/G4 was 1.8 times lower than that of Py/CsF/G4, as confirmed by ICP-MS, it exhibited almost one tenth conductivity compared to Py/Cs/G4. Moreover, although An/CsF/G4 (3.4 V) possessed a wider stability window than that of Py/CsF/G4 (2.6 V), the discharge and charge reactions of BiF_3 did not progress in An/CsF/G4. On the other hand, promising first discharge and especially charge capacities were obtained in Py/CsF/G4. The reason for these two different behaviors was associated with the higher Lewis acidity of the borate in DiOB-Py than in DiOB-An. The higher CsF dissolution based on ICP-MS and diffusivity of their fluoride anion complexes (based on the ionic conductivity measurement) in Py/CsF/G4 confirm this approach. In addition, not only high reversible capacity but also lower discharge/charge polarization were achieved in Py/CsF/G4 compare to the similar state-of-the-art electrolyte in the literature. Moreover, we found that the electrochemical and chemical dissolution of the active material was very low or did not occur in Py/CsF/G4. Thus, the direct defluorination/fluorination mechanism in Py/CsF/G4 is considered to occur more effectively. Therefore, (borolan-2-yl)pyridine - based borate (DiOB-Py) has imparted high fluoride ion conductivity and CsF solubility and thereby allowed a successful fluoride shuttle-based redox reaction.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

5. Acknowledgments

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